Output:-

Que :- Consider a disk queue with requests for l/O to blocks on cylinders 98, I83, 41, 122, 14,124,65,67. The SSTF scheduling algorithm is used. The head is initially at cylinder number 53. The cylinders are numbered from 0 to 199. Write a Program to calculate the total head movement (in number of cylinders) incurred while servicing these requests.

Code :-

```
  GNU nano 7.2                                                          sstf.c *
#include <stdio.h>
#include <stdlib.h>

int main() {
    int req[] = {98,183,41,122,14,124,65,67};
    int n = 8;
    int head = 53;
    int visited[8] = {0};
    int total = 0;

    printf("Disk Request Sequence: ");
    for(int i = 0; i < n; i++)
        printf("%d ", req[i]);

    printf("\nInitial Head Position: %d\n", head);

    for(int i = 0; i < n; i++) {
        int min = 9999, index = -1;

        for(int j = 0; j < n; j++) {
            if(!visited[j]) {
                int dist = abs(head - req[j]);
                if(dist < min) {
                    min = dist;
                    index = j;
                }
            }
        }

        total += min;
        head = req[index];
        visited[index] = 1;
    }

    printf("\nTotal Head Movement = %d\n", total);

    return 0;
}
```

Output:-

```
bhushan123@LAPTOP-0V4BCI
bhushan123@LAPTOP-0V4BCL4Q:~$ nano sstf.c
bhushan123@LAPTOP-0V4BCL4Q:~$ gcc sstf.c -o sstf
bhushan123@LAPTOP-0V4BCL4Q:~$ ./sstf
Total Head Movement = 323
bhushan123@LAPTOP-0V4BCL4Q:~$
```